

Carbon Monoxide on Purpose

A bovine oxygen carrier that arrives loaded with a famous poison, and the reasoning that says this is on purpose

There is a product in the oxygen-carrier family whose central design feature will sound, on first hearing, like a mistake. It is a bovine hemoglobin, chemically wrapped in polyethylene glycol, and it is manufactured and stored with carbon monoxide already bound to its iron. Carbon monoxide is the gas that kills people in garages and closed rooms precisely because it takes the place of oxygen on hemoglobin. A medicine that begins by filling every binding site with carbon monoxide seems to be built backward.

The product is known as Sanguinate, a PEGylated carboxyhemoglobin from bovine blood, developed by Prolong Pharmaceuticals. Reading how its developers justify the design is one of the more instructive exercises in this field, because it forces a rethinking of what an oxygen carrier is for. This essay walks through the argument, states clearly which parts are supported and which are hopeful, and draws the lessons that matter for patients following the wider BHOC story.

The gas we mistake for only a poison

Start with a fact that surprises many people. Your body makes carbon monoxide continuously. An enzyme called heme oxygenase breaks down heme, the iron-containing ring from old hemoglobin, and one of the products is carbon monoxide, along with iron and a green-yellow pigment called biliverdin. This is not a rare or pathological event. Every day a healthy adult breaks down a substantial quantity of hemoglobin and generates a small, steady flow of the gas.

Over the past few decades, physiologists have shown that this endogenous carbon monoxide behaves as a signaling molecule. It relaxes blood vessels, dampens certain inflammatory responses, and influences how cells handle stress. It belongs to a small group of gaseous transmitters, sometimes called gasotransmitters, along with nitric oxide and hydrogen sulfide. The three share a history: each was known first as a poison, and each was later found to be a messenger the body makes on purpose. Readers of another post in this series will recognize hydrogen sulfide from the deep-sea tubeworm, which uses a protein to carry it safely.

The step from that science to a drug is a small one conceptually. If carbon monoxide at low doses can calm inflamed, oxygen-starved tissue, then a delivery system for small, controlled amounts of the gas might help. Chemists developed several classes of carbon monoxide-releasing molecules to test the idea. The Sanguinate design takes a different route: it uses hemoglobin, the body's own carbon monoxide carrier, as the delivery vehicle.

Why carboxyhemoglobin is a good storage form

There is a second, more prosaic reason to load hemoglobin with carbon monoxide, and it is pure chemistry. Free hemoglobin is fragile. In the oxygenated state it slowly loses an electron from its iron and turns into methemoglobin, which cannot carry oxygen and which sheds toxic

heme. The oxygenated form is the most susceptible. In the deoxygenated state the iron is reactive in other ways, including toward nitric oxide.

Carbon monoxide bound to hemoglobin protects the iron. The carboxy form is reported to be considerably more resistant to oxidation than the oxy form, because carbon monoxide binds tightly and holds the iron in its ferrous state. A manufacturer who stores the product as carboxyhemoglobin therefore has a molecule that stays useful on the shelf. When the product is infused, the carbon monoxide is gradually displaced by oxygen in the lung and in tissue, since oxygen at high concentration can compete with carbon monoxide, and the released carbon monoxide can act as a signal. The gas is at once a preservative and a payload.

Those are two independent justifications, one from stability chemistry and one from pharmacology, that point to the same molecule. When two independent arguments converge, it is worth paying attention, though convergence is not proof that either is right.

Wrapping the molecule

The second element of the design is the polyethylene glycol. PEGylation attaches long, water-loving chains to the protein surface. It has been used for decades to extend the circulation time of protein drugs and to shield them from immune recognition. In the context of a hemoglobin carrier, the chains do several things at once. They enlarge the molecule's effective size in solution, which reduces leakage through the kidney. They add a hydration shell that increases viscosity and colloid osmotic pressure without requiring a high concentration of protein. And, as the developers argue, they lower the molecule's tendency to trigger vasoconstriction, which in earlier products was linked to the scavenging of nitric oxide in the vessel wall.

PEGylation also changes oxygen affinity. Reports on this class of PEGylated hemoglobins describe a high-affinity carrier, with a P50 well below that of normal blood. That is the opposite of the bovine glutamer products discussed in another post on this site. A high-affinity design is consistent with the microvascular hypothesis: a carrier that holds oxygen tightly delivers it only where tissue is severely hypoxic, and does not trigger arteriolar constriction upstream. The developers presented this as an advantage in conditions such as sickle cell crisis and ischemia, where blood vessels have narrowed and red cells cannot reach the tissue.

Three gases, one protein

It helps to line up the three gasotransmitters against the single protein that meets all of them, because the comparison shows why hemoglobin is both the problem and the opportunity.

Nitric oxide is made by the vessel lining and diffuses a short distance to the smooth muscle around it, where it relaxes the muscle and widens the vessel. Free hemoglobin in plasma is an avid scavenger of nitric oxide, which is the root of the vasoconstriction that undermined earlier carriers. The problem is that hemoglobin removes a gas the body wants to keep.

Carbon monoxide is made in small amounts by heme breakdown, binds hemoglobin about two hundred times more tightly than oxygen does, and relaxes vessels by a different route, through the same enzyme that nitric oxide activates and through potassium channels. Here the protein holds a gas that the body may want to release. Loading it in advance turns the scavenger into a depot.

Hydrogen sulfide is made by several enzymes, is toxic at high levels, and in the tubeworm is carried on zinc sites of the protein without harming the heme. It shows that a hemoglobin can be engineered, by evolution, to carry a second gas safely.

Seen together, the three gases suggest a design vocabulary. A future carrier might scavenge less nitric oxide, deliver a dose of carbon monoxide, and shield its iron from sulfide. That is speculative. It is also a coherent research program, and the PEGylated carboxy product is its first tangible example.

A thought experiment about dose

Because the safety argument depends on dose, a numerical thought experiment is helpful. The numbers below are illustrative only and are not drawn from any trial.

Suppose a patient has five liters of blood with a total hemoglobin of ten grams per deciliter, so fifty grams circulating. Suppose an infusion adds three grams of a carboxy carrier, six percent of the total. If every molecule of the carrier were saturated with carbon monoxide, then in the first moments the carboxy fraction of the patient's total hemoglobin would rise by about six percentage points from the carrier alone. In practice the carbon monoxide begins to exchange almost at once with oxygen in the lung, so the peak would be lower and the fraction would decline over hours.

Compare this with a heavy smoker, who can have baseline carboxyhemoglobin of five to ten percent, and with a person in a garage, in whom levels above twenty or thirty percent produce headache, confusion and collapse. Under this admittedly rough arithmetic, a small dose sits in the range that a smoker tolerates daily, and a very large one would approach the range of clinical concern. That is why the protocol details matter, and why the honest summary has to include the phrase depends on the dose.

The reason to go through this is not to reassure. It is to show the kind of reasoning a reader can apply to any claim about safety. Ask for the dose, ask what fraction of hemoglobin it represents, and ask how that compares with a familiar exposure. Numbers turn a scary word into a quantity.

The clinical ambitions

The published descriptions of the product's development list a set of targets that reads like a map of the places where oxygen fails locally rather than globally. The company obtained an orphan drug designation from the United States regulator for sickle cell disease. The designation was reported to be based on what the manufacturer described as the product's ability to reverse sickling in laboratory settings, and it was cited in connection with complications including vaso-occlusive crisis, acute chest syndrome, leg ulcers and stroke. Trials were reported in other oxygen-deprivation conditions, including delayed cerebral ischemia after bleeding around the brain and delayed function in kidney grafts from donors. Early trials were also reported in beta-thalassemia outside the United States.

There is also the use that made the product visible outside the specialty literature: emergency compassionate use for patients with life-threatening anemia who refused blood. These single-patient cases, like others in this field, generated case reports rather than controlled data.

I want to be exact about what I know and do not know. I have read the developers' summary papers and the regulator's public designation notice. I have not seen complete, peer-reviewed results of a randomized trial showing that the product improves a clinical outcome in any of the listed conditions, and I will not imply otherwise. Where a phase 2 trial has been run, the responsible response is to consult the trial report itself. My purpose in this essay is to examine the logic of the design, not to certify its success.

What the sickle cell rationale actually is

Sickle cell disease is worth a moment because its logic is a good test of the whole idea. In sickle cell disease a mutation in the beta chain makes deoxygenated hemoglobin polymerize inside the red cell, deforming it into a rigid crescent that jams small vessels. The jam causes a local oxygen deficit, which promotes more deoxygenation, which promotes more sickling. It is a feedback loop.

The argument for a cell-free, high-affinity carrier is that it can slip past the blockage. A red cell is a large, stiff object when sickled and cannot squeeze through a partly closed vessel. A dissolved polymer or a PEGylated protein moves with the plasma, and plasma still flows through narrow channels that cells cannot enter. If the protein delivers even a little oxygen to the tissue beyond the blockage, it might interrupt the loop. The addition of carbon monoxide may contribute an anti-inflammatory and vessel-relaxing effect that eases the vessel narrowing.

This is a beautiful hypothesis. It is also a hypothesis, and a hard one to test, because sickle cell crises are unpredictable, patient reports of pain are subjective, and the natural course of a crisis is to improve on its own. Trials in this area are difficult to design and easy to misread. A modest positive signal in a small trial may be noise. A negative one may reflect the wrong dose or the wrong timing. That does not make the idea wrong. It means the honest summary is: interesting mechanism, limited proof.

The measurement puzzle

Anyone who has read the earlier post on measurement will see a difficulty at once. If a product is loaded with carbon monoxide, the instruments we use to read oxygenation become unreliable in a specific way.

A standard finger pulse oximeter, which uses two wavelengths of light, cannot distinguish oxyhemoglobin from carboxyhemoglobin; both absorb red light similarly, so it reads a falsely reassuring saturation when carbon monoxide is present. This is the same reason the instrument fails in real carbon monoxide poisoning. A laboratory co-oximeter, which measures at many wavelengths, can separate the two species, and so any trial of a carboxyhemoglobin product has to specify that arterial samples be analyzed on a co-oximeter and that finger readings not be relied on.

Here is the rub. The total carboxyhemoglobin level in the blood after an infusion of a carboxyhemoglobin product is a mixture of the product's own carbon monoxide and the patient's native levels, and the interpretation depends on the dose. A small dose adds a fraction of a percent to the total carboxyhemoglobin fraction. A large one could raise it to a level that in a smoker or a poisoned patient would prompt concern. A careful protocol therefore has to

define dose limits in terms of the carbon monoxide delivered, monitor carboxyhemoglobin by co-oximetry, and educate staff to disregard the pulse oximeter's number. Any of these steps can fail in a busy emergency department, and that is a real operational hazard.

The safety question, asked fairly

Is it safe to give carbon monoxide to a patient? The pharmacologist's answer is that dose determines the poison, as with most drugs. Carbon monoxide is dangerous when it displaces a large fraction of the body's oxygen-carrying capacity, typically levels of carboxyhemoglobin well into the double digits. The amount delivered by a modest infusion of a carboxyhemoglobin carrier is a fraction of that, and it is diluted into a patient who is receiving less than a full replacement of their hemoglobin.

But the pharmacologist's answer is incomplete for three reasons.

First, the patients most likely to receive the product are already compromised. A sickle cell patient in crisis, a patient with a brain bleed or a patient with a failing organ may have less reserve to tolerate any reduction in usable hemoglobin. Second, carbon monoxide binds not only to hemoglobin but also to other heme proteins, including cytochromes in mitochondria, and its net effect on cellular respiration depends on dose in ways that are still being mapped. Third, carbon monoxide has a long history as an occupational and environmental hazard, and the public and clinicians carry a strong prior against it. A hospital pharmacy committee may struggle to approve a drug whose active ingredient is a poison by common understanding, even if the pharmacology is sound.

None of these objections is decisive. All of them are the kind of question a regulatory reviewer would ask, and the public documents suggest the developers took them seriously by monitoring carboxyhemoglobin levels in their early trials. I will simply state that a careful reader should treat the safety case as plausible but incompletely established.

What the design teaches about the whole field

Step back from the product itself, because I think its importance is broader than any one program.

The mainstream story of oxygen carriers is a story about substitution: a molecule replaces a red cell, and success is measured by whether it carries as much oxygen as blood. The carbon monoxide design proposes a different category. The molecule is a dual-purpose object, an oxygen transporter that is also a drug. It exploits the fact that hemoglobin is the natural carrier of gaseous signaling molecules as well as of oxygen, and it treats that as a feature.

If the idea is right, it opens a path for the whole family. A carrier could be loaded with a therapeutic amount of a gas, or with nitric oxide in a protected form to counter the scavenging problem, or with a stabilizing ligand that extends shelf life. The bovine glutamers discussed in another post carry oxygen and nothing else. A future generation might carry oxygen and a dose of signal.

If the idea is wrong, it teaches something too. It would mean the vessel-relaxing and anti-inflammatory actions of carbon monoxide at these doses are too weak to matter, and that the

protective value of the PEG shield and the high affinity did the work. Either way, the comparison of PEGylated carboxy products with unmodified bovine polymers offers a natural experiment: two designs with related chemistry and very different philosophies. A well-organized field would design its trials to tell them apart.

A word for patient communities

The lesson for advocates is largely about vocabulary. When someone tells you a product is a bovine hemoglobin oxygen carrier, ask which kind. The category contains molecules that differ in size, in oxygen affinity, in what they carry besides oxygen, and in what disease they aim to treat. A bovine glutamer of average mass around two hundred fifty kilodaltons with a mid-range affinity is not the same object as a PEGylated carboxyhemoglobin with a high affinity. Lumping them together, as public discussion often does, makes evidence for one seem to apply to the other, and makes the failure of one seem to condemn both.

It is equally important to be honest about the difference between mechanism and outcome. A design can be clever and still fail. The history of the field is littered with clever designs. What earns trust is a patient, transparent record of trial results, and until that record exists the appropriate stance toward any of these products is curiosity, not conviction.

The poison bound to that iron is not a paradox. It is a proposition, stated in chemistry: that the body's oldest toxin, given in a controlled dose and carried by its own transport protein, might be an ally. The proposition has been made. The test is the trial data.

Further reading

The developers' summary of mechanism and clinical experience appears in an Artificial Organs paper from 2017 on PEGylated carboxyhemoglobin bovine. The regulator's orphan designation for sickle cell disease was reported in 2015. Background on carbon monoxide as an endogenous signaling molecule, produced by heme oxygenase, is available in general physiology and pharmacology reviews. Readers should consult trial registries and peer-reviewed reports for current clinical results, which I have not attempted to summarize.